

AGRON 746 – Environmental Soil Physics

Mid-term Exam 2 Example Questions

The problems for exam 2 will be taken from this list with some modifications. Read carefully. If you need to access soil parameters to solve these problems, use Table 2 in the manuscript by Rawls et al 1983.

1. Using Darcy's law as a starting point, write the Green-Ampt equation for ponded infiltration rate and define each variable in the equation. Create a sketch representing each component of the equation and highlight which variables remain constant and which variables change over time. List at least two assumptions made when using this equation. Note that this question does not involve any calculation.
2. Why do we often observe a distinct wetting front when water infiltrates into soil that was relatively dry prior to infiltration?
3. Ponded infiltration is occurring in a loam soil. The ponding depth is 5.0 cm. The wetting front is now 15. cm deep. The initial water content of the soil was $0.10 \text{ m}^3 \text{ m}^{-3}$.
 - a. Estimate the current infiltration rate.
 - b. Estimate the cumulative infiltration.
4. Rainfall infiltration is occurring in a sandy loam soil. The rainfall rate is constant at 1.0 inch per hour. The initial water content was $0.15 \text{ m}^3 \text{ m}^{-3}$.
 - a. Estimate the depth of the wetting front when ponding first occurs.
 - b. Estimate how much water infiltrates before ponding occurs.
 - c. Estimate the time from the start of the rain until ponding.
5. Estimate the soil water content at which drainage from the bottom of the soil profile equals 0.50 mm d^{-1} for a deeply wetted, uniform, sandy loam. Use the soil properties from the handout titled "Overview of the Campbell soil hydraulic property functions." Assume a saturated water content of $0.45 \text{ cm}^3 \text{ cm}^{-3}$.
6. What are the three main stages of soil erosion? For erosion by water, describe the primary processes or factors involved with each of these three stages.
7. Define rainfall interception and list two factors that influence its magnitude.
8. Define soil crusts. Describe the effects soil crusts have on infiltration, seedling emergence, and water erosion.
9. Describe three mechanisms of crust formation. Explain why irrigation water with high sodium adsorption ratio may increase the risk of soil crust formation.
10. What is the typical range of raindrop sizes? Why?
11. Before a 1.00 inch rainfall event, the soil water content was $0.15 \text{ m}^3 \text{ m}^{-3}$ in the top 20.0 cm of soil. After the event, the soil water content was $0.25 \text{ m}^3 \text{ m}^{-3}$ in the top 20.0 cm of soil. Assume there were no losses except interception. What percentage of the rainfall was intercepted?
12. Define rainfall erosivity and specify three factors which influence it.
13. Define soil erodibility and specify three factors which influence it.
14. Define soil splash and describe its relationship to shear stress and soil strength.

15. A pulse of sediment in runoff water is delivered to a Tuttle Creek Lake. How long will it take for soil particles in the silt range with diameter of 5×10^{-3} mm to settle to a depth of 15 m (approximate depth of the lake)? Assume:

$$\eta = 1 \times 10^{-3} \text{ kg m}^{-1} \text{ s}^{-1}$$

$$\rho_s = 2650 \text{ kg m}^{-3}$$

$$\rho_f = 1000 \text{ kg m}^{-3}$$

$$g = 9.81 \text{ m s}^{-2}$$